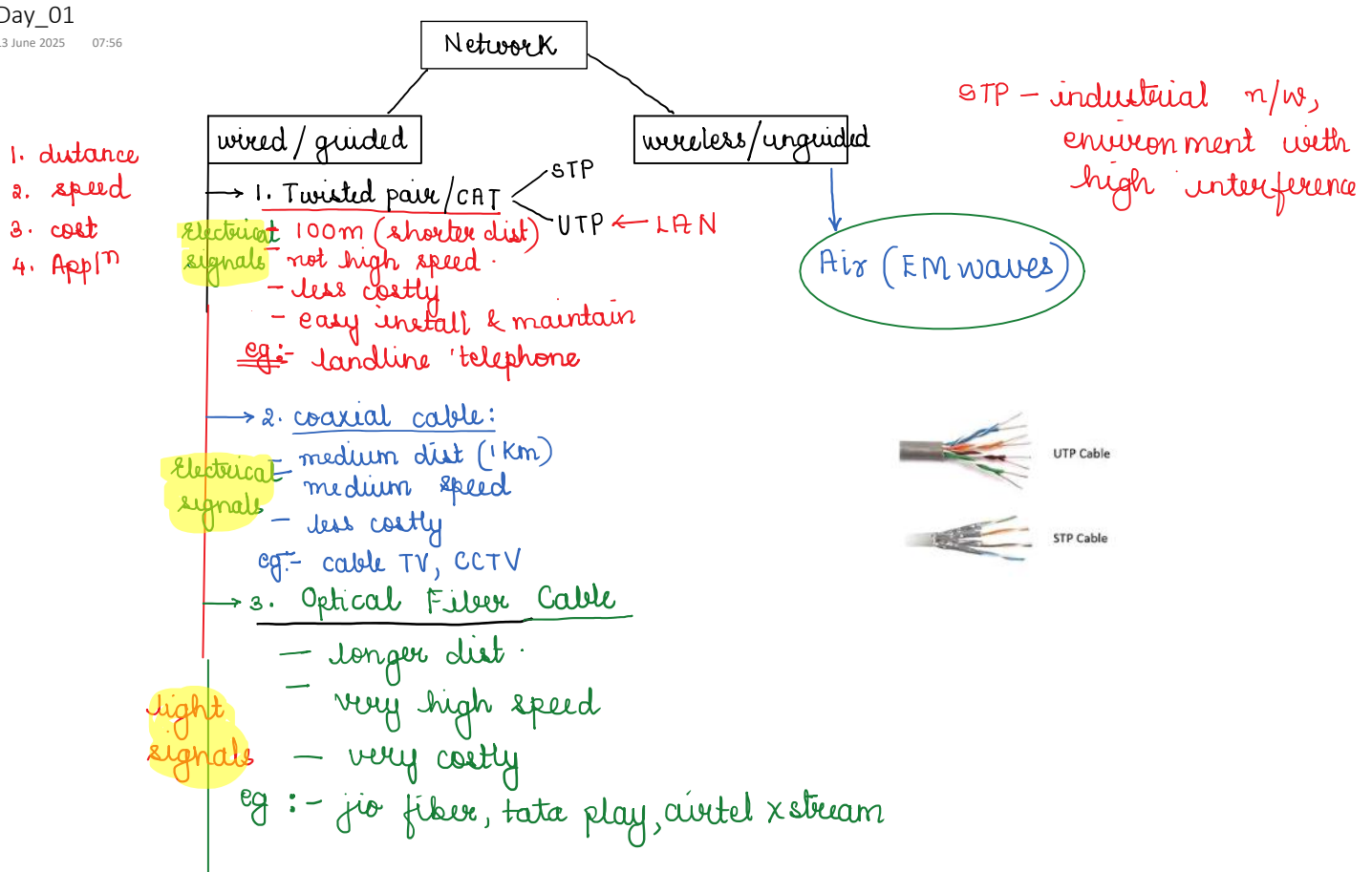




Transmission Mode (Data flow direction)

1. Simple Mode:-

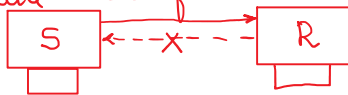
data transfers only in one directⁿ



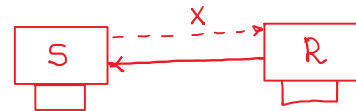
eg:- keyboard, TV remote, pager, radio

2. Half Duplex Mode:-

data transfers in both direction but not at the same time

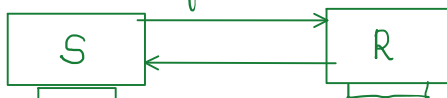


eg:- walkie talkie, FAX, etc



3. Full Duplex Mode:-

data transfers in both direction simultaneously.

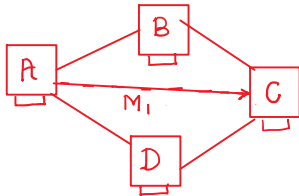


eg:- mobile, telephone, video, voice, chat, etc.

Switching

selectⁿ of best route to transmit data across the networks.

1. Circuit Switching:-



CAM₁

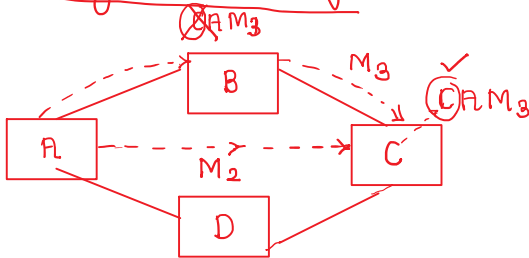
C = destination
A = source
M₁ = message

eg:- telephone

- Fixed bandwidth
- consistent quality
- Not efficient for heavy traffic/collision.
- security

1. connection established
2. data transferred
3. connection terminated

2. Message Switching



CAM₂

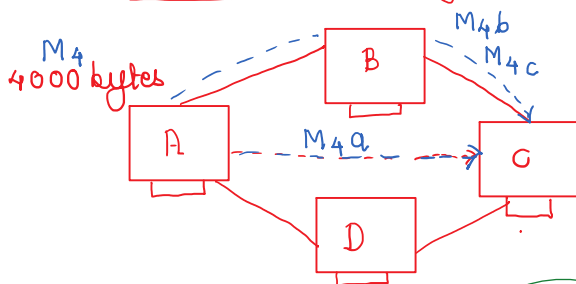
CAM₃

store & forward technique

eg:- SMS

1. no need of dedicated path
2. high delay in the msg being transmitted
3. connectionless
4. not reliable

3. Packet Switching:-



ethernet packet

1500 bytes

ethernet technology

- 1) each packet is treated independently.
- 2) no such dedicated path.

M₄ = 4000 bytes

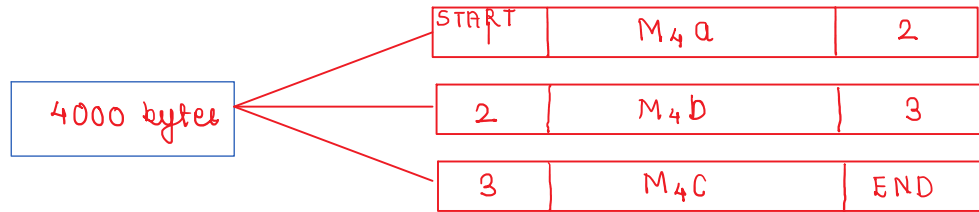
M_{4a} = 1500 bytes

M_{4b} = 1500 bytes

M_{4c} = 1000 bytes

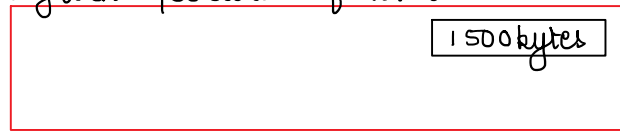
1500 bytes

1 data packet = 1500 bytes



eg: VoLTE → Voice over Long Term Evolution

Bandwidth → max capacity of a wired/wireless n/w to transmit max data in a given period of time.

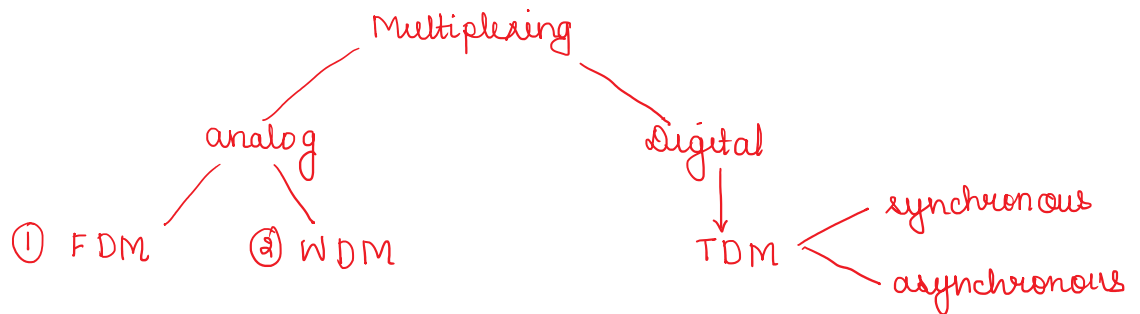


100 mbps

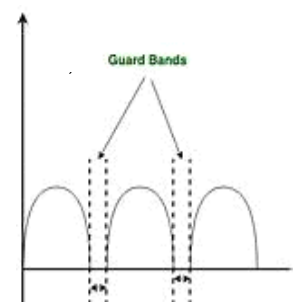
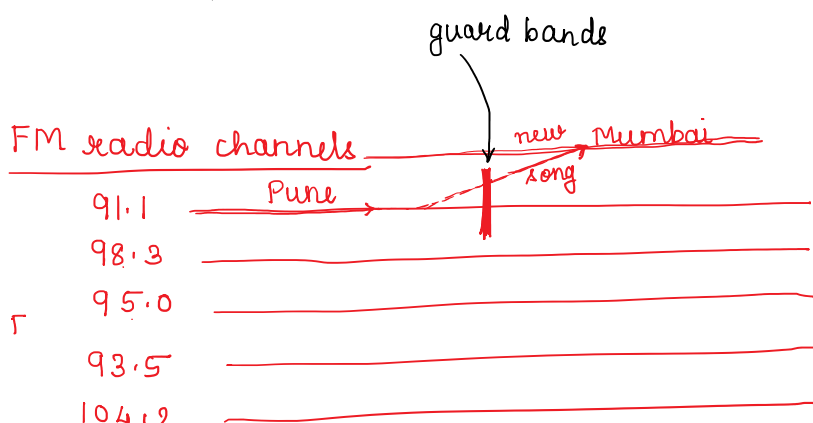
overcome wastage of bandwidth we use multiplexing

mux → convert multiple signals into one signal ← (multiplexing)

demux → convert one signal into multiple signals ← (demultiplexing)



FDM

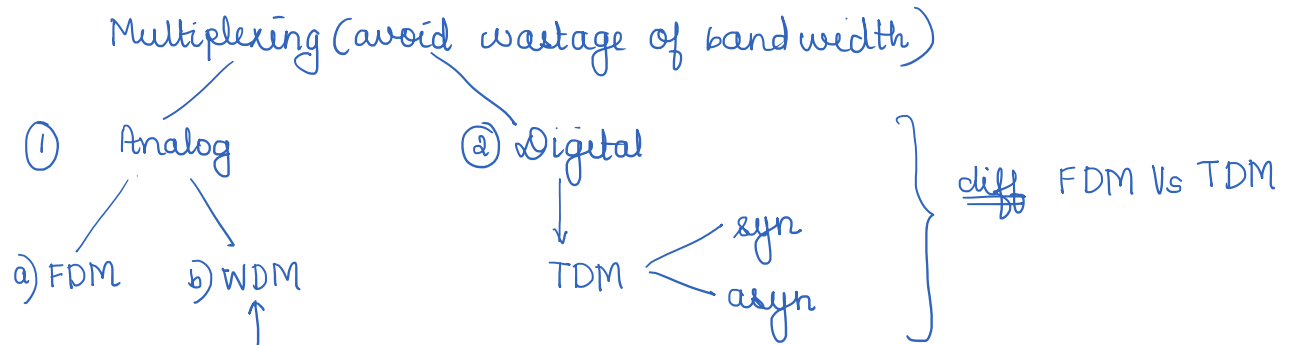
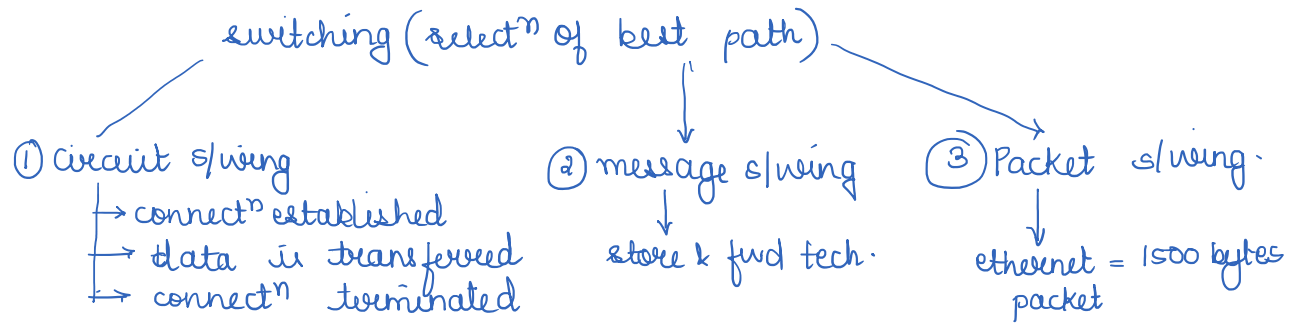


105.0



(3) Full Duplex mode

- ① simplex mode
- ② Half duplex mode
- ③ Full Duplex mode



MAC addr^r → 6 bytes (48 bits)

